govern and co-ordinate the instinctive powers, the much smaller mass of medulla oblongata is evidently the chief centre of the powers themselves, and without its preservation, motion, sensation, and even life itself cease to animate the organism.

There are twelve pairs of nerves given off from the brain, and from thence to the organs of special sense, and different portions of the head.

The first pair are called the olfactory nerves, and terminate in the inner portions of the nose.

The second pair are called the optic nerves, and supply the retina of the eye.

The third, fourth, fifth, and sixth pairs are also distributed about the eyes; the third, fourth, and sixth pairs being motor nerves.

The fifth pair, called trifacial, are sensory, and send off branches which supply the cheeks, nose, and mouth.

The seventh pair are the general motor nerves of the facial muscles.

The eighth pair, called auditory nerves, end in the interior of the eye.

The ninth pair supply the back of the mouth and pharynx.

The tenth pair, the important pneumogastric nerves, originating in the medulla oblongata, communicate with the lungs, heart, and stomach, and ramify through the entire viscera to a considerable extent. (See Fig. 3.)

The eleventh pair supply the muscles of the neck.

The twelfth pair communicate with the tongue and the organs of speech.

The two pairs, eleventh and twelfth, are called